SSC MTS 2024 question paper with answers

1 October 2024 · Shift 1

78 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 12 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 13 questions

Q.2

Ravi goes to college from his village at the speed of 5km/h and returns at the speed of 4km/h. If he takes 9 hours in all, find the distance (in km) between the village and the college.

(A) 18
(B) 20
(C) 9
(D) 40

Q.3

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 7% per year rate of interest for 5 years.

(A) 915
(B) 815
(C) 835
(D) 875

Q.4

A borrows ₹89,500 for 3 years at 7% p.a. simple interest. B borrows the same amount at simple interest of 9% for the first year, 8% for the second year, and 7% for the third year. What is the total simple interest (in ₹) paid by A and B?

(A) 42,075
(B) 40,275
(C) 47,025
(D) 45,027

Q.5

Two trains running towards each other on parallel tracks, at speeds of 45 km/hr and 51 km/hr, take 21 seconds to cross each other. If the length of one train is 290 m, then the length (in m) of the other train is:

(A) 250
(B) 230
(C) 290
(D) 270

Q.7

The average weight of a certain number of students in a group was 72 kg. If 11 students, with an average weight of 76 kg each, leave the group, and 5 students, each with an average weight of 79 kg, join the group, the average weight of the students in the group decreases by 0.6 kg. The number of students initially in the group was:

(A) 19
(B) 25
(C) 21
(D) 17

Q.8

The average of 9 numbers is 3. If the average of the first 8 of these numbers is 2, the 9th number is:

(A) 15
(B) 14
(C) 12
(D) 11

Q.12
The ratio of the present ages of Ram and Shyam is 3 : 4, and 4 years ago the ratio was 2 : 3. What will be the ratio of the ages of Ram and Shyam 4 years from now?

(A) 1 : 5
(B) 2 : 5
(C) 4 : 5
(D) 3 : 5

Q.13
Find the area of the quadrilateral ABCD, the coordinates of whose vertices are A(6,-3),B(-4,-2),C(3,1) and D(5,0)?

(A) 30 unit^2
(B) 16 unit^2
(C) 25 unit^2
(D) 21 unit^2

Q.14
Find the average of 300, 129, 49, 120 and 102.

(A) 120
(B) 130
(C) 140
(D) 150

Q.15
A cylindrical disc of thickness 1 cm is made by melting two solid cylinders of radii 4 cm and 5 cm, and length 6 cm and 4 cm, respectively. The radius of the new cylinder is:

(A) 21 cm
(B) 7 cm
(C) 14 cm
(D) 28 cm

Q.17
By selling an article for ₹20, a dealer makes a profit of 10%. By how much should he increase his selling price (in ₹, rounded off to two decimal place) so as to make a profit of 20%?

(A) 3.82
(B) 1.82
(C) 4.82
(D) 2.82

Q.19
What sum (in ₹) will earn a simple interest of ₹240 in 4 years at 12% per year rate of interest?

(A) 750
(B) 500
(C) 250
(D) 1000

Q.20
The least number of square tiles required to pave the ceiling of a room 525 centimetre long and 378 centimetre broad is:

(A) 450
(B) 150
(C) 178
(D) 380

Reasoning Ability and Problem Solving 15 questions

Q.1
What should come in place of the question mark (?) in the given series? 53, 52, 39, 38, ?, 24

(A) 25
(B) 26
(C) 28
(D) 23

Q.2
What should come in place of the question mark (?) in the given series based on the English alphabetical order? TKU, UNP, VQK, WTF, ?

(A) WAW
(B) XWA
(C) YXB
(D) XXA

Q.3

GY 16 is related to DV 23 in a certain way. In the same way, OD 11 is related to LA 18. To which of the following is LW 13 related, following the same logic?

(A) HT 19
(B) IT 20
(C) IS 20
(D) HS 20

Q.4

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some carrots are radishes. Some cucumbers are radishes. Conclusions: I. All cucumbers are carrots. II. Some cucumbers are carrots.

(A) Only conclusion II follows
(B) Only conclusion I follows
(C) Both conclusions I and II follow
(D) Neither conclusion I nor II follows

Q.5

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (12, 4, 160) (6, 11, 157)

(A) (9, 15, 90)
(B) (14, 7, 245)
(C) (6, 15, 90)
(D) (21, 4, 168)

Q.6

P, Q, R, S, T and U are sitting around a circular table facing the centre (but not necessarily in the same order). P is second to the left of T. Q is an immediate neighbour of both U and P. S and P are immediate neighbours. Who among the following has both T and U as his immediate neighbours?

(A) P
(B) R
(C) Q
(D) S

Q.8

In a certain code language, ‘TALE’ is coded as ‘5173’ and ‘AIDS’ is coded as ‘6452’.What is the code for ‘A’ in the given code language?

(A) 2
(B) 5
(C) 6
(D) 1

Q.10

Which of the following interchanges of signs would make the given equation correct? $15 - 54 \times 18 + 16 \div 2 = 44$

(A) + and ×
(B) – and +
(C) ÷ and ×
(D) – and ÷

Q.11

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All parrots are eagles. No parrot is a crow. Conclusions: (I) No eagle is a crow. (II) All crows are parrots.

(A) Only conclusion I follows
(B) Neither conclusion I nor II follows
(C) Both conclusions I and II follow
(D) Only conclusion II follows

Q.12

In a certain code language, 'EAGER' is writtenas ‘CGGTG’ and 'LEARN' is writtenas ‘GNAPT’. How will 'HINDI' be written in that language?

(A) JKNKF
(B) KJNKF
(C) KJNFK
(D) KJPKF

Q.13

18 is related to 80 following a certain logic. Following the same logic, 23 is related to 100. To which ofthe following is 36 related, following the same logic?(NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed)

(A) 144
(B) 152
(C) 142
(D) 138

Q.14

UBIP is related to YFMT in a certain way based on the English alphabetical order. In the same way, IPWD is related to MTAH. To which of the following is OVCJ related, following the same logic?

(A) SZGN
(B) SMGJ
(C) SZMG
(D) SJZM

Q.15

In a certain code language, ‘PLAY’ is coded as ‘4831’ and ‘CLAY’ is coded as ‘1824’. What is the code for ‘C’ in that language?

(A) 4
(B) 2
(C) 8
(D) 1

Q.16

Seven people A, B, C, D, E, F and G are sitting around a circular table, facing the center (but not necessarily in the same order). G is sitting third to the left of E. A is sitting to the immediate left of G. D is the immediate neighbour of A and B. F is sitting to the immediate left of C. How many people are sitting between F and D when counted from the right of F?

(A) Three
(B) Four
(C) Five
(D) Two

Q.17

What will come in the place of the question mark (?) in the following equation if ‘–’ and ‘×’ are interchanged and ‘÷’ and ‘+’ are interchanged? 6 – 8 ÷ 12 + 3 × 9 + 3 ÷ 17 = ?

(A) 71
(B) 66
(C) 56
(D) 2

General Awareness 25 questions

Q.1

After which of the following Acts did India come to be governed in the name of the British crown?

(A) Charter Act of 1833
(B) Charter Act of 1853
(C) Government of India Act, 1858
(D) Government of India Act, 1919

Q.2

Who was the Indian match referee in the ICC - T20 World Cup 2021 tournament?

(A) Venkatesh Prasad
(B) Sunil Gavaskar
(C) Rahul Dravid
(D) Javagal Srinath

Q.3

In general, deficit is considered as a _____ concept.

(A) flow
(B) surplus
(C) margin
(D) stock

Q.4

Which of the following dances of Goa is performed to celebrate the victory of the Maratha warriors over the Portuguese?

(A) Goff
(B) Morulem
(C) Zemmado
(D) Ghodemodni

Q.5

What is the approx width of broad-gauge railway network in India?

(A) 1.3 metres
(B) 1.2 metres
(C) 1.6 metres
(D) 1 metre

Q.6

A coastal area was flooded with sea water and resulted in heavy salt deposits in the nearby area. After a few days, it was observed that many plants died. Based on these statements, which of the following is the most probable cause of the plant deaths? A. Plants died due to infection from sea water borne germs. B. Plants died due to absence of sunlight. C. Plants died due to plasmolysis in the plant cells. D. Plants died because they were eaten by sea organisms.

(A) A
(B) B
(C) C
(D) D

Q.7

The Universal Adult Franchise in Indian Constitution is applicable to Indian Citizens from the age of _____.

(A) 22 years
(B) 17 years
(C) 18 years
(D) 21 years

Q.8

The PM-SVANidhi Scheme (2020) is run by which Ministry of the Government of India?

(A) Ministry of Rural Development
(B) Ministry of Housing and Urban Affairs
(C) Ministry of Finance
(D) Ministry of Labour and Employment

Q.9

In which stadium did Indian cricketer Saurav Ganguly take off and waive his t-shirt?

(A) The Melbourne Cricket Ground, Australia
(B) Cape Town Stadium, South Africa
(C) Perth Stadium, Australia
(D) Lord's Stadium, England

Q.10

Who was appointed as the Viceroy of India in place of Lord Wavell in February 1947?

(A) Lord Irwin
(B) Lord Linlithgow
(C) Lord Willingdon
(D) Lord Mountbatten

Q.11
Which of the following age groups is considered as working population or working age in India?

(A) 25-65 years
(B) 15-40 years
(C) 15-59 years
(D) 0-60 years

Q.12
The Bidar Fort of Karnataka is irregularly _____ in shape.

(A) rectangular
(B) circular
(C) square
(D) rhomboid

Q.13
When was National Judicial Commission Bill, which is a private member bill to regulate the appointment of judges through the National Judicial Commission, introduced in Rajya Sabha?

(A) August 2023
(B) July 2023
(C) March 2022
(D) December 2022

Q.14
Which musical duo is part of the Khayal singing lineage of the Banaras Gharana?

(A) Salamat Ali and Nazakat Ali Khan
(B) Rajan and Sajan Mishra
(C) Subhan Kasim and Subhan Bab
(D) Sandhya and Prasun Banerjee

Q.15
Who among the following deciphered the Brahmi Script of the Mauryan period?

(A) Rakhal Das Banerji
(B) Herbert Hope Risley
(C) James Prinsep
(D) James Mill

Q.16
According to Indian Meteorology, which of the following is NOT a season experienced in India?

(A) The hot weather season
(B) The southwest monsoon season
(C) Equatorial season
(D) The cold weather season

Q.17
Which of the following can be a basis of difference between a salamander and a house-wall lizard apart from their habitat?

(A) Scales on the skin
(B) Three-chambered heart
(C) Cold-blood
(D) Egg-laying feature

Q.18
The IUPAC name of CH_3-$(CH_2)_{18}$-CH_3 is _____.

(A) icosane
(B) dodecane
(C) decane
(D) triacontane

Q.19
A party recognised by the Election Commission of India that secures at least _____ of the total votes in the General Election to Lok Sabha or Legislative Assembly in four or more states in addition to winning four Lok Sabha seats is called a National Party.

(A) eight per cent
(B) one per cent
(C) six per cent
(D) three per cent

Q.20

Hornbill festival is organised every year in which of the following states?

(A) Goa
(B) Sikkim
(C) Tripura
(D) Nagaland

Q.21

Which of the following novels is NOT written by Aravind Adiga?

(A) Selection Day
(B) Last Man in Tower
(C) The English Teacher
(D) The White Tiger

Q.22

The Mughal emperor, Babur was a devotee of the _____Sufi Silsila.

(A) Qadri
(B) Chisti
(C) Suhrawardi
(D) Naqshbandi

Q.23

Bawris are traditionally used for:

(A) getting petroleum oil
(B) collecting garbage
(C) getting useful minerals
(D) storing and recharging water

Q.24

Which of the following Kathak dancers received the Padma Shri of 2022?

(A) Kamalini and Nalini Asthana
(B) Shovana Narayan
(C) Sunayana Hazarlal Agrawal
(D) Shakti and Mukti Mohan

Q.25

At the Sansad Ratna Awards 2023, who among the following received the Dr. APJ Abdul Kalam Lifetime Achievement Award?

(A) HD Deve Gowda
(B) Laxminarayan Yadav
(C) Nagendra Singh
(D) TK Rangarajan

English Language and Comprehension 25 questions

Q.1

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. He dropped out / at his friend's house / on his way home / the day before yesterday.

(A) at his friend's house
(B) on his way home
(C) He dropped out
(D) the day before yesterday

Q.2

Select the most appropriate expression that can substitute the underlined word in the given sentence. Sonali is “contemplating” whether to accept the offer from Saanvi or not; she is still not in a position to be able to arrive at a decision.

(A) chewing the cud
(B) blowing hot and cold
(C) getting her act together
(D) raising the eyebrows

Q.3

The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don’t find any error, mark ‘No error’ as your answer. I am / cleaning the sofa / when she called me.

(A) I am
(B) cleaning the sofa
(C) No error
(D) when she called me

Q.4

Select the correctly spelt word to fill in the blank. Arvind wanted to bake a cake for Riya's birthday, so he mixed the _____, eggs and butter.

(A) flawer
(B) flover
(C) flower
(D) flour

Q.5

Select the option that rectifies the spelling error in the given sentence. The student received an A+ for her exellent performance.

(A) perfarmence
(B) performence
(C) excelleant
(D) excellent

Q.6

Select the most appropriate option to fill in the blank. The _____ of the scientific experiment raised questions about the validity of the results.

(A) transparency
(B) verbosity
(C) ambiguity
(D) meticulousness

Q.7

Select the most appropriate synonym of the given word. Inspire

(A) Reach
(B) Aspire
(C) Stimulate
(D) Daunt

Q.8

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. The teacher checked / whether each of them / have a blanket / as it was a cold night.

(A) have a blanket
(B) whether each of them
(C) as it was a cold night
(D) The teacher checked

Q.9

The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don’t find any error, mark ‘No error’ as your answer. She is / as more intelligent as / her brothers are.

(A) her brothers are.
(B) as more intelligent as
(C) She is
(D) No error.

Q.10

Select the INCORRECTLY spelt word.

(A) tentative
(B) perepharal
(C) belated
(D) stringent

Q.11

Select the most appropriate option that can substitute the underlined segment in the given sentence. Raghav, who “studies tourism” at present, is planning to go abroad after his graduation.

(A) is studying
(B) will study
(C) study
(D) studied

Q.12

Select the word that has the same meaning (SYNONYM) as the underlined word in the given sentence. The disposal of “hazardous” waste in hospitals and factories is a serious problem and needs to be taken care of urgently.

(A) belligerent
(B) dangerous
(C) hazy
(D) benign

Q.13

Select the most appropriate option to fill in the blank. The student leader has _____ politics recently.

(A) gathered
(B) jumped
(C) approached
(D) entered

Q.14

Select the INCORRECTLY spelt word from the given sentence. The gardener's meticulous care ensured the vibrent colours of the flowers.

(A) Gardener's
(B) Vibrent
(C) Care
(D) Meticulous

Q.15

Select the most appropriate option to fill in the blank. Please put the dishes in the _____ after you finish eating.

(A) sink
(B) sync
(C) shrink
(D) cinque

Q.16

You overhear someone say, "He's always on cloud nine." What does this idiom mean in the given context?

(A) Always busy
(B) Extremely happy
(C) Often absent
(D) Frequently daydreaming

Q.17

Select the most appropriate synonym of ‘scanty’ in the sentence given below. India has varied regions which have excess, moderate and deficient rainfall all through the year.

(A) Excess
(B) Moderate
(C) Varied
(D) Deficient

Q.18

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. The society / of the / our place / is good.

(A) The society
(B) is good
(C) our place
(D) of the

Q.19

Select the present perfect tense of the verb given in the brackets to fill in the blank. Alex _____ (study) two programming languages.

(A) have study
(B) has studied
(C) had studied
(D) has been studying

Q.20

Select the most appropriate option that can substitute the underlined segment in the sentence. She was deeply “effected by” the news, to the point of tears.

(A) affected by
(B) effected from
(C) affect from
(D) affected of

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. It was one January morning, very early — a pinching, frosty morning — the cove all grey ____ 1 ____ hoar-frost and the ripple lapping softly on the stones. The captain had _____ 2____ earlier than usual and set out down the beach. Mother was upstairs with father and I was _____ 3____ the breakfast table against the captain's return when the parlour door opened and a man stepped in on whom I had never set my eyes before. He was a pale, tallowy creature, wanting two fingers of the left hand, and _____ 4____ he wore a cutlass, he did not look much like a fighter. I had always my eye open for seafaring _____ 5_____ with one leg or two, and I remember this one puzzled me.

Select the most appropriate option to fill in blank number 1.

(A) among
(B) amongst
(C) with
(D) about

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. It was one January morning, very early — a pinching, frosty morning — the cove all grey ____ 1 ____ hoar-frost and the ripple lapping softly on the stones. The captain had _____ 2____ earlier than usual and set out down the beach. Mother was upstairs with father and I was _____ 3____ the breakfast table against the captain's return when the parlour door opened and a man stepped in on whom I had never set my eyes before. He was a pale, tallowy creature, wanting two fingers of the left hand, and _____ 4____ he wore a cutlass, he did not look much like a fighter. I had always my eye open for seafaring _____ 5_____ with one leg or two, and I remember this one puzzled me.

Select the most appropriate option to fill in blank number 2.

(A) written
(B) waltzed
(C) risen
(D) danced

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. It was one January morning, very early — a pinching, frosty morning — the cove all grey ____ 1 ____ hoar-frost and the ripple lapping softly on the stones. The captain had _____ 2____ earlier than usual and set out down the beach. Mother was upstairs with father and I was _____ 3____ the breakfast table against the captain's return when the parlour door opened and a man stepped in on whom I had never set my eyes before. He was a pale, tallowy creature, wanting two fingers of the left hand, and _____ 4____ he wore a cutlass, he did not look much like a fighter. I had always my eye open for seafaring _____ 5_____ with one leg or two, and I remember this one puzzled me.

Select the most appropriate option to fill in blank number 3.

(A) throwing
(B) laying
(C) playing
(D) lying

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. It was one January morning, very early — a pinching, frosty morning — the cove all grey ____ 1 ____ hoar-frost and the ripple lapping softly on the stones. The captain had _____ 2____ earlier than usual and set out down the beach. Mother was upstairs with father and I was _____ 3____ the breakfast table against the captain's return when the parlour door opened and a man stepped in on whom I had never set my eyes before. He was a pale, tallowy creature, wanting two fingers of the left hand, and _____ 4____ he wore a cutlass, he did not look much like a fighter. I had always my eye open for seafaring _____ 5_____ with one leg or two, and I remember this one puzzled me.

Select the most appropriate option to fill in blank number 4.

(A) throw
(B) through
(C) threw
(D) though

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. It was one January morning, very early — a pinching, frosty morning — the cove all grey ____ 1 ____ hoar-frost and the ripple lapping softly on the stones. The captain had _____ 2____ earlier than usual and set out down the beach. Mother was upstairs with father and I was _____ 3____ the breakfast table against the captain's return when the parlour door opened and a man stepped in on whom I had never set my eyes before. He was a pale, tallowy creature, wanting two fingers of the left hand, and _____ 4____ he wore a cutlass, he did not look much like a fighter. I had always my eye open for seafaring _____ 5_____ with one leg or two, and I remember this one puzzled me.

Select the most appropriate option to fill in blank number 5.

(A) birds
(B) men
(C) plants
(D) flowers

Answer key

Numerical and Mathematical Ability

Q.2 **(B)**	Q.3 **(D)**	Q.4 **(B)**	Q.5 **(D)**	Q.7 **(C)**	Q.8 **(D)**
Q.12 **(C)**	Q.13 **(D)**	Q.14 **(C)**	Q.15 **(C)**	Q.17 **(B)**	Q.19 **(B)**
Q.20 **(A)**					

Reasoning Ability and Problem Solving

Q.1 **(A)**	Q.2 **(B)**	Q.3 **(B)**	Q.4 **(D)**	Q.5 **(B)**	Q.6 **(B)**
Q.8 **(B)**	Q.10 **(C)**	Q.11 **(B)**	Q.12 **(B)**	Q.13 **(B)**	Q.14 **(A)**
Q.15 **(B)**	Q.16 **(A)**	Q.17 **(B)**			

General Awareness

Q.1 **(C)**	Q.2 **(D)**	Q.3 **(A)**	Q.4 **(D)**	Q.5 **(C)**	Q.6 **(C)**
Q.7 **(C)**	Q.8 **(B)**	Q.9 **(D)**	Q.10 **(D)**	Q.11 **(C)**	Q.12 **(D)**
Q.13 **(D)**	Q.14 **(B)**	Q.15 **(C)**	Q.16 **(C)**	Q.17 **(A)**	Q.18 **(A)**
Q.19 **(C)**	Q.20 **(D)**	Q.21 **(C)**	Q.22 **(D)**	Q.23 **(D)**	Q.24 **(A)**
Q.25 **(D)**					

English Language and Comprehension

Q.1 **(C)**	Q.2 **(A)**	Q.3 **(A)**	Q.4 **(D)**	Q.5 **(D)**	Q.6 **(C)**
Q.7 **(C)**	Q.8 **(A)**	Q.9 **(B)**	Q.10 **(B)**	Q.11 **(A)**	Q.12 **(B)**
Q.13 **(D)**	Q.14 **(B)**	Q.15 **(A)**	Q.16 **(B)**	Q.17 **(D)**	Q.18 **(D)**
Q.19 **(B)**	Q.20 **(A)**	Q.21 **(C)**	Q.22 **(C)**	Q.23 **(B)**	Q.24 **(D)**
Q.25 **(B)**					

Explanations

Numerical and Mathematical Ability

Q.2 · Answer (B) 20

Time = d/5 + d/4 = 9d/20 = 9 hours, so d = 20 km.

Q.3 · Answer (D) 875

SI = 2500 × 7 × 5/100 = ₹875.

Q.4 · Answer (B) 40,275

A pays 7 × 3 = 21% of ₹89,500 = ₹18,795 and B pays 9 + 8 + 7 = 24% of it = ₹21,480. Together ₹40,275.

Q.5 · Answer (D) 270

Moving towards each other, the relative speed is 45 + 51 = 96 km/h = 80/3 m/s, so in 21 seconds they cover 560 m, the sum of their lengths. The other train is 560 − 290 = 270 m.

Q.7 · Answer (C) 21

With n students at first, the new total is 72n − 11 × 76 + 5 × 79 = 72n − 441 for n − 6 students averaging 71.4 kg. 71.4(n − 6) = 72n − 441 gives 0.6n = 12.6, so n = 21.

Q.8 · Answer (D) 11

The 9 numbers total 9 × 3 = 27 and the first 8 total 8 × 2 = 16, so the 9th is 27 − 16 = 11.

Q.12 · Answer (C) 4 : 5

Let the ages be 3x and 4x: (3x − 4)/(4x − 4) = 2/3 gives 9x − 12 = 8x − 8, so x = 4 and the ages are 12 and 16. In 4 years: 16 : 20 = 4 : 5.

Q.13 · Answer (D) 21 unit²

By the shoelace formula, area = ½|(6 × −2 − (−4) × −3) + (−4 × 1 − 3 × −2) + (3 × 0 − 5 × 1) + (5 × −3 − 6 × 0)| = ½|−24 + 2 − 5 − 15| = ½ × 42 = 21 square units.

Q.14 · Answer (C) 140

(300 + 129 + 49 + 120 + 102)/5 = 700/5 = 140.

Q.15 · Answer (C) 14 cm

The volume is unchanged: π(4² × 6 + 5² × 4) = 196π cm³. A disc 1 cm thick has volume πr² × 1, so r² = 196 and r = 14 cm.

Q.17 · Answer (B) 1.82

Cost price = 20/1.1 = ₹18.18. A 20% profit needs a selling price of 1.2 × 18.18 = ₹21.82, an increase of ₹1.82.

Q.19 · Answer (B) 500

P = SI × 100/(R × T) = 240 × 100/(12 × 4) = ₹500.

Q.20 · Answer (A) 450

The largest square tile has side HCF(525, 378) = 21 cm, so (525/21) × (378/21) = 25 × 18 = 450 tiles.

Reasoning Ability and Problem Solving

Q.1 · Answer (A) 25

The terms fall by 1 and 13 alternately: 53 − 1 = 52, 52 − 13 = 39, 39 − 1 = 38, 38 − 13 = 25 and 25 − 1 = 24.

Q.2 · Answer (B) XWA

First letters move +1 (T, U, V, W, X), second letters +3 (K, N, Q, T, W) and third letters −5 (U, P, K, F, A): XWA.

Q.3 · Answer (B) IT 20

Each letter moves 3 places back and the number rises by 7: G → D, Y → V, 16 → 23. So L → I, W → T and 13 → 20: IT 20.

Q.4 · Answer (D) Neither conclusion I nor II follows
Carrots and cucumbers each overlap radishes, but nothing links carrots with cucumbers, so neither 'all' nor 'some cucumbers are carrots' is certain.

Q.5 · Answer (B) (14, 7, 245)
The third number is the sum of the squares of the first two: $12^2 + 4^2 = 144 + 16 = 160$ and $6^2 + 11^2 = 36 + 121 = 157$. Likewise $14^2 + 7^2 = 196 + 49 = 245$.

Q.6 · Answer (B) R
Going clockwise (to the left, for people facing the centre): T, S, P, Q, U, R. P is second to T's left; Q sits between P and U, so S takes the seat between T and P, and R the last seat, between U and T.

Q.8 · Answer (B) 5
A is the only letter in both words and 5 the only digit in both codes, so A = 5.

Q.10 · Answer (C) ÷ and ×
Interchanging ÷ and ×: 15 – 54 ÷ 18 + 16 × 2 = 15 – 3 + 32 = 44.

Q.11 · Answer (B) Neither conclusion I nor II follows
No parrot is a crow, so no crow is a parrot and II is false. Eagles that are not parrots may or may not be crows, so I does not follow either.

Q.12 · Answer (B) KJNKF
The first two letters swap, the middle one stays and the last two swap; then each swapped letter moves 2 places forward: EAGER → AEGRE → CGGTG and LEARN → ELANR → GNAPT. So HINDI → IHNID → KJNKF.

Q.13 · Answer (B) 152
Add 2 and multiply by 4: (18 + 2) × 4 = 80 and (23 + 2) × 4 = 100. So (36 + 2) × 4 = 152.

Q.14 · Answer (A) SZGN
Each letter moves 4 places forward: U → Y, B → F, I → M, P → T, and W → A, wrapping round. So OVCJ → SZGN.

Q.15 · Answer (B) 2
L, A and Y are in both words, as are 8, 1 and 4 in both codes. The letter left in CLAY is C and the digit left in its code is 2, so C = 2.

Q.16 · Answer (A) Three
Going clockwise (to the left, for people facing the centre) from E: E, C, F, G, A, D, B. Counting to F's right from F, C, E and B come before D: three people.

Q.17 · Answer (B) 66
After interchanging the signs: 6 × 8 + 12 ÷ 3 – 9 ÷ 3 + 17 = 48 + 4 – 3 + 17 = 66.

General Awareness

Q.1 · Answer (C) Government of India Act, 1858
The Government of India Act, 1858, passed after the revolt of 1857, ended the East India Company's rule: India was thereafter governed by and in the name of the British Crown.

Q.2 · Answer (D) Javagal Srinath
Javagal Srinath, India's member of the ICC Elite Panel of Match Referees, was one of the four match referees at the 2021 T20 World Cup.

Q.3 · Answer (A) flow
A deficit is measured over a period, such as a year, so it is a flow; debt, the deficits accumulated up to a point in time, is a stock.

Q.4 · Answer (D) Ghodemodni
Ghode Modni, a horse dance performed at Shigmo by men wearing horse frames and carrying swords, commemorates the victory of the Maratha warriors (the Ranes of Sattari) over the Portuguese.

Q.5 · Answer (C) 1.6 metres
India's broad gauge is 1.676 m (5 ft 6 in) between the rails; of the options, 1.6 metres is the closest.

Q.6 · Answer (C) C
The salt made the soil water more concentrated than the cell sap, so water left the root cells by osmosis and their cytoplasm shrank away from the cell wall: plasmolysis.

Q.7 · Answer (C) 18 years
Under Article 326, as amended by the 61st Amendment (1988), every citizen aged 18 or over may vote; the age was 21 before.

Q.8 · Answer (B) Ministry of Housing and Urban Affairs
PM SVANidhi, the micro-credit scheme for street vendors launched in June 2020, is run by the Ministry of Housing and Urban Affairs.

Q.9 · Answer (D) Lord’s Stadium, England
Ganguly took off his shirt and waved it from the Lord's balcony after India won the NatWest Series final against England in July 2002.

Q.10 · Answer (D) Lord Mountbatten
In February 1947 Attlee announced that Lord Mountbatten would succeed Lord Wavell as Viceroy; he took office in March 1947.

Q.11 · Answer (C) 15-59 years
India's census counts the population aged 15-59 years as the working-age group (international usage often takes 15-64).

Q.12 · Answer (D) rhomboid
Bidar Fort, built by the Bahmani sultan Ahmad Shah I from 1429, is laid out as an irregular rhomboid on the edge of the Bidar plateau.

Q.13 · Answer (D) December 2022
The National Judicial Commission Bill, 2022, a private member's bill moved by CPI(M) MP Bikash Ranjan Bhattacharya, was introduced in the Rajya Sabha on 9 December 2022.

Q.14 · Answer (B) Rajan and Sajan Mishra
The brothers Rajan and Sajan Mishra were khayal singers of the Banaras gharana.

Q.15 · Answer (C) James Prinsep
James Prinsep, an officer of the East India Company's mint, deciphered Brahmi in 1837, which made Asoka's inscriptions readable.

Q.16 · Answer (C) Equatorial season
Indian meteorology recognises four seasons: the cold weather season, the hot weather season, the southwest monsoon season and the retreating monsoon season. There is no 'equatorial season'.

Q.17 · Answer (A) Scales on the skin
The house lizard is a reptile with dry, scaly skin; the salamander is an amphibian with moist skin and no scales. Both are cold-blooded, lay eggs and have three-chambered hearts.

Q.18 · Answer (A) icosane
The chain has 1 + 18 + 1 = 20 carbon atoms, and the alkane with 20 carbons is icosane (eicosane); dodecane has 12, decane 10 and triacontane 30.

Q.19 · Answer (C) six per cent
Under the Election Symbols Order, 1968, one way to become a National Party is to win at least 6% of the valid votes in four or more states at a Lok Sabha or assembly election and also win four Lok Sabha seats.

Q.20 · Answer (D) Nagaland
Nagaland holds the Hornbill Festival every year from 1 to 10 December at Kisama, near Kohima.

Q.21 · Answer (C) The English Teacher
'The English Teacher' (1945) is by R. K. Narayan. 'The White Tiger', 'Last Man in Tower' and 'Selection Day' are Aravind Adiga's.

Q.22 · Answer (D) Naqshbandi
Babur, like the Timurids before him, followed the Naqshbandi order and revered its Central Asian master Khwaja Ubaidullah Ahrar.

Q.23 · Answer (D) storing and recharging water
Bawris (baolis) are stepwells, traditionally used to store rainwater and recharge groundwater, especially in Rajasthan.

Q.24 · Answer (A) Kamalini and Nalini Asthana
The Kathak sisters Kamalini and Nalini Asthana of Agra received the Padma Shri together in 2022.

Q.25 · Answer (D) TK Rangarajan
The Dr APJ Abdul Kalam Lifetime Achievement Award at the Sansad Ratna Awards 2023 went to T. K. Rangarajan, a former two-term Rajya Sabha member of the CPI(M).

English Language and Comprehension

Q.1 · Answer (C) He dropped out
To 'drop out' is to withdraw; a casual visit to a friend's house is 'dropped in' (or 'dropped by').

Q.2 · Answer (A) chewing the cud
To 'chew the cud' is to think something over slowly and carefully, as Sonali is doing.

Q.3 · Answer (A) I am
An action in progress when something else happened takes the past continuous: 'I was cleaning the sofa when she called me.'

Q.4 · Answer (D) flour
A cake is made with 'flour', ground grain; 'flower' sounds the same but means a bloom.

Q.5 · Answer (D) excellent
The correct spelling is 'excellent'.

Q.6 · Answer (C) ambiguity
'Ambiguity', being open to more than one reading, is what would cast doubt on the results; transparency and meticulousness would support them.

Q.7 · Answer (C) Stimulate
To 'inspire' is to stimulate someone to feel or act; 'daunt' is nearly its opposite.

Q.8 · Answer (A) have a blanket
'Each of them' is singular and takes a singular verb: 'has (or had) a blanket', not 'have'.

Q.9 · Answer (B) as more intelligent as
Equality is shown with 'as … as' and the positive degree: 'She is as intelligent as her brothers are.'

Q.10 · Answer (B) perepharal
The correct spelling is 'peripheral'.

Q.11 · Answer (A) is studying
An action going on around the present takes the present continuous: 'who is studying tourism at present'.

Q.12 · Answer (B) dangerous
'Hazardous' means risky or harmful: 'dangerous'. 'Benign' is its opposite.

Q.13 · Answer (D) entered
One 'enters' politics, that is, takes it up.

Q.14 · Answer (B) Vibrent
The correct spelling is 'vibrant'.

Q.15 · Answer (A) sink
Dirty dishes go in the 'sink'; 'sync' and 'cinque' only sound similar.

Q.16 · Answer (B) Extremely happy
To be 'on cloud nine' is to be extremely happy.

Q.17 · Answer (D) Deficient
'Scanty' means less than is needed: 'deficient', as in deficient rainfall.

Q.18 · Answer (D) of the
'The' and 'our' cannot both stand before 'place': 'The society of our place is good.'

Q.19 · Answer (B) has studied
The present perfect is 'has' or 'have' with the past participle; with the singular 'Alex' it is 'has studied'.

Q.20 · Answer (A) affected by
'Affect' is the verb meaning to influence or move, and one is 'affected by' something: 'deeply affected by the news'. As a verb, 'effect' means to bring about.

Q.21 · Answer (C) with
The cove was 'grey with hoar-frost': 'with' names what covers it and makes it grey, as in 'white with snow'.

Q.22 · Answer (C) risen
'The captain had risen earlier than usual': 'risen' is the past participle of 'rise', to get up.

Q.23 · Answer (B) laying
To 'lay' the table is to set it, and 'lay' takes an object: 'I was laying the breakfast table'. 'Lying', from 'lie', takes no object.

Q.24 · Answer (D) though
'Though' introduces a contrast: though he wore a cutlass, he did not look much like a fighter.

Q.25 · Answer (B) men
'Seafaring men' are sailors, and only people fit 'with one leg or two'; birds, plants and flowers do not.

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